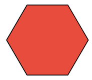
A396138 – smallest polyhex carrying a complete n-coloring

Each panel shows a minimal connected polyhex realizing a(n), with its cells colored by a complete n-coloring (every one of the C(n,2) color pairs occurs on some cell-cell edge).

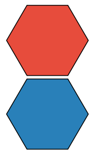
a(1..10) = 1, 2, 3, 5, 7, 9, 12, 16, 18, 21

Computed by Peter Exley, June 2026

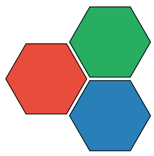
$a(1) = 1$ **[PROVED]** 1 colors



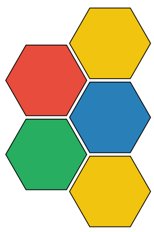
$a(2) = 2$ **[PROVED]** 2 colors



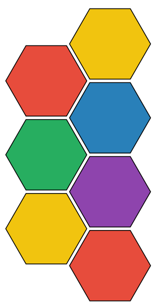
$a(3) = 3$ **[PROVED]** 3 colors



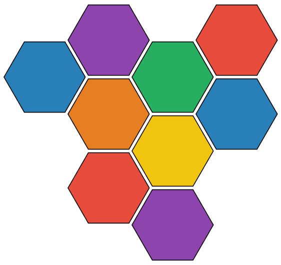
$a(4) = 5$ **[PROVED]** 4 colors



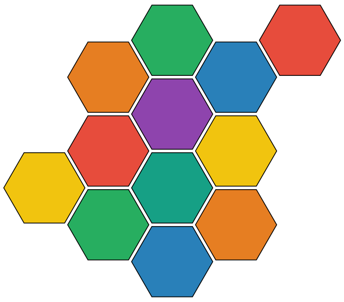
$a(5) = 7$ **[PROVED]** 5 colors



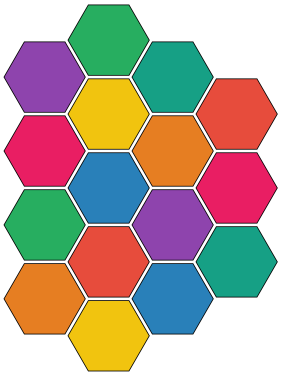
$a(6) = 9$ **[PROVED]** 6 colors



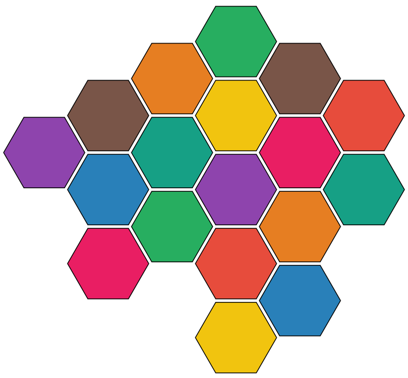
$a(7) = 12$ **[PROVED]** 7 colors



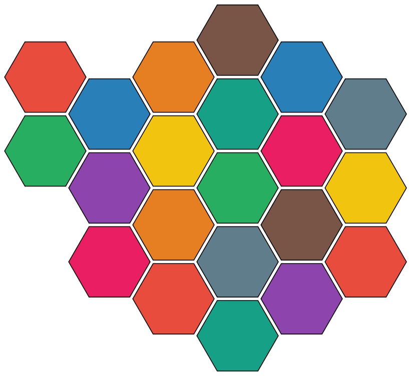
$a(8) = 16$ **[PROVED]** 8 colors



$a(9) = 18$ **[PROVED]** 9 colors



$a(10) = 21$ **[PROVED]** 10 colors



Color Legend

A	B	C	D	E	F	G
H	I	J				